# Motivation

- If the problem implies a physical context of several dimensions, a multidimensional array may be used.
  - Chairs in a classroom
  - Rooms on different floors of a building
  - Coordinates on a map
  - Coordinates in space
  - A timetable (hours versus days)

# 2D Arrays

- In a classroom where the chairs are organized as a grid of 8 rows and 10 columns

```
int i, j, chair[8][10];
for (i=0; i<8; i++)
    for (j=0; j<10; j++)
        scanf("%d",&chair[i][j]);
```

# Example #1

- Construct a student's timetable. Read course code (assume all courses have distinct codes).

```
int table[8][5];
for (i=0; i<8; i++)
    for (j=0; j<5; j++)
        scanf("%d",&table[i][j]);
```

# Example #2

- Store a map of every pixel on the screen (256 colors/pixel). Assume a resolution of 1024x768.

```
unsigned char screen[1024][768];
```

# Example #3

- Read the number of inhabitants in every flat in a building. Assume there are 10 floors with 5 flats in every floor. Find the flats that have occupancy above the average.

# Example #3

```
int flat[10][5], i, j, sum=0;
float avg;

for (i=0; i<10; i++)
   for (j=0; j<5; j++)
   {  scanf("%d", &flat[i][j]);
      sum += flat[i][j];
   }
avg = sum/50.0;
for (i=0; i<10; i++)
   for (j=0; j<5; j++)
      if (flat[i][j]>avg)
         printf("On floor %d, in flat %d\n",i,j);
```

# Example #4

- Mark every soldier with "1" on a map. Rest is all zeros.
- Find the coordinates of all soldiers that can reach a given coordinate in 10 steps.

# Example #4

```
#define ABS(x) ((x)<0) ? -(x) : (x)
int map[1000][1000], int coord_x, coord_y, i, j;

for (i=0; i<1000; i++)
   for (j=0; j<1000; j++)
      scanf("%d", &map[i][j]);

scanf("%d %d", &coord_x, &coord_y); /* Read coordinates*/

for (i=coord_x-10; i<=coord_x+10; i++)
   for (j=coord_y-10; j<=coord_y+10; j++)
      if (map[i][j])
         if ((ABS(coord_x-i)+ABS(coord_y-j) <= 10)
            printf("%d  %d", i, j);
```

# Example #5

- Find the number of cell phones in the coverage of a base station. Assume cell radius is 20 units.

# Example #5

```
#define SQR(x) (x)*(x)
int map[1000][1000], int BS_x, BS_y, i, j, count=0;

for (i=0; i<1000; i++)
   for (j=0; j<1000; j++)
      scanf("%d", &map[i][j]);

scanf("%d %d", &BS_x, &BS_y); /* BS coordinates*/

for (i=BS_x-20; i<=BS_x+20; i++)
   for (j=BS_y-20; j<=coord_y+20; j++)
      if (SQR(BS_x-i)+SQR(BS_y-j) <= 400)
         count++;
printf("%d cell phones in the cell\n", count);
```

# 3D Array

- Store the day-of-the-week info for all days in for three years.

```
enum day_of_week {SUN,MON,TUE,WED,THU,FRI,SAT,SUN};
enum month {JAN,FEB,MAR,APR,MAY,JUN,JUL,AUG,SEP,OCT,NOV,DEC};

enum day_of_week day[3][12][31];

day[0][MAY][19]=MON;
```

# 4D Arrays

- As ATC, you want to check if the route of a plane is valid.

```
#define T 100
#define X 200
#define Y 200
#define Z 100
int space[T][X][Y][Z];
...
for (t=0; t<T; t++)
{  scanf("%d %d %d", &x, &y, &z);
   if (space[t][x][y][z])
      printf("There is an other plane at the coordinates"
             "(%d,%d,%d) at time %d\n", x,y,z,t);
}
```

# More Dimensions

- Store the grade from each question in each exam for each course of all students.

```
int question[1000][40][3][5];
```

# More Dimensions

- Calculate the monthly salary of each worker. 10TL/hr.

```
int work[100][12][31][24];
int worker, month, day, hour;
float salary;

... /* Initialize work array */

for (worker=0; worker<100; worker++)
{  salary=0;
   for (month=0; month<12; month++)
      for (day=0; day<31; day++)
         for (hour=0; hour<24; hour++)
            salary += 10.0*work[worker][month][day][hour]/60.0;
   printf("Salary for Month #%d is %f\n", month, salary);
}
```